



Certyfikat PMP: Project Scheduling Tools

The Project Manager's Master Blueprint
for Time & Resource Control

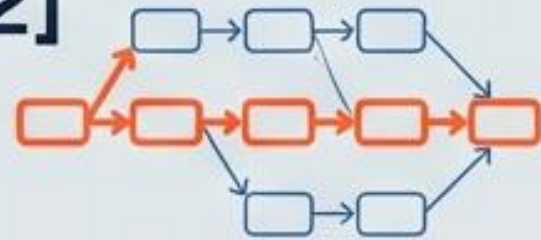
dr Zbigniew Galar



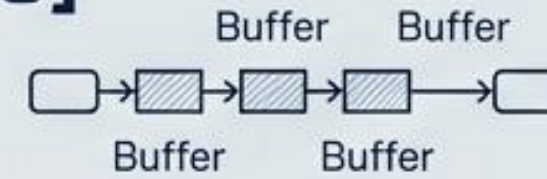
The Develop Schedule Control Panel

[1]

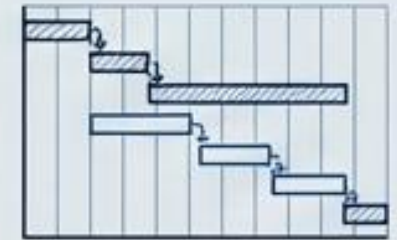
Schedule Network Analysis

[2]

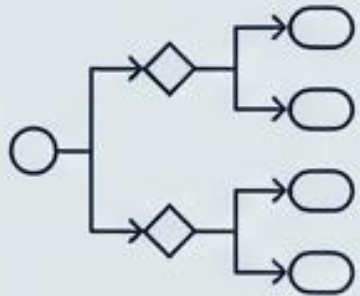
Critical Path Method (CPM)

[3]

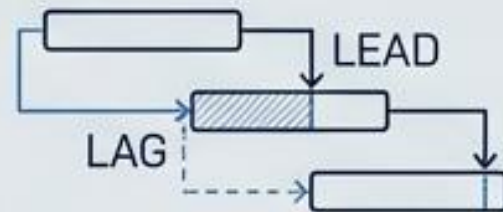
Critical Chain Method

[4]

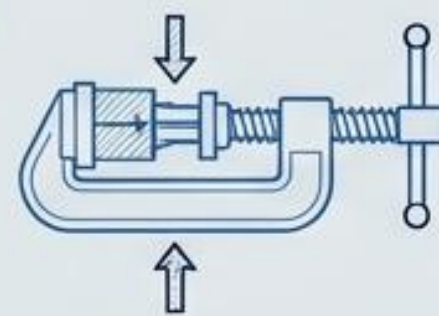
Resource Optimization Techniques

[5]

Modeling Techniques

[6]

Leads and Lags

[7]

Schedule Compression

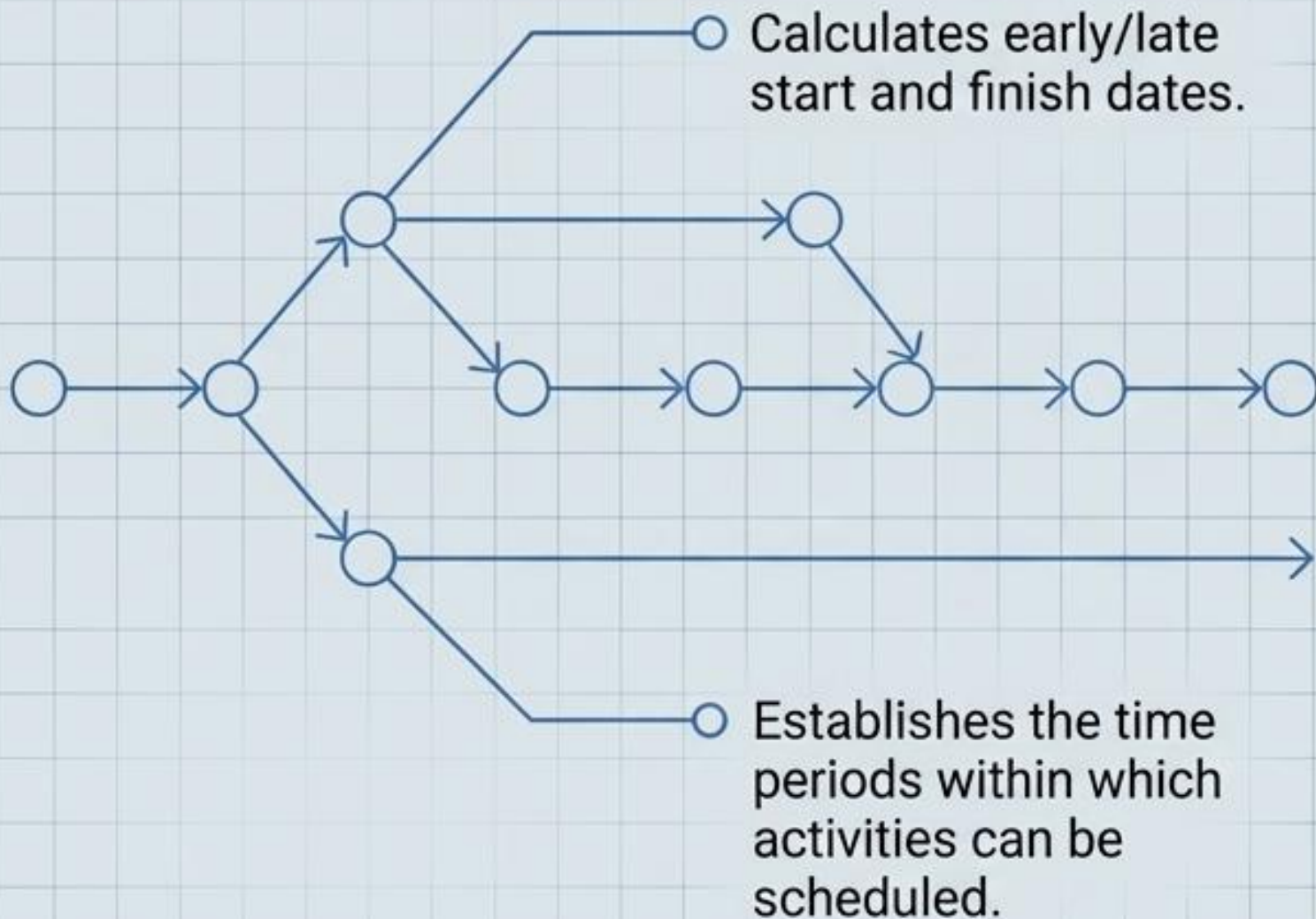
[8]

Scheduling Tool

These tools sequence logically—from establishing a theoretical baseline (CPM) to accounting for risk (PERT), adjusting for real-world constraints (Optimization), and executing emergency corrections (Compression).

Establishing the Theoretical Baseline

THE FUNCTION



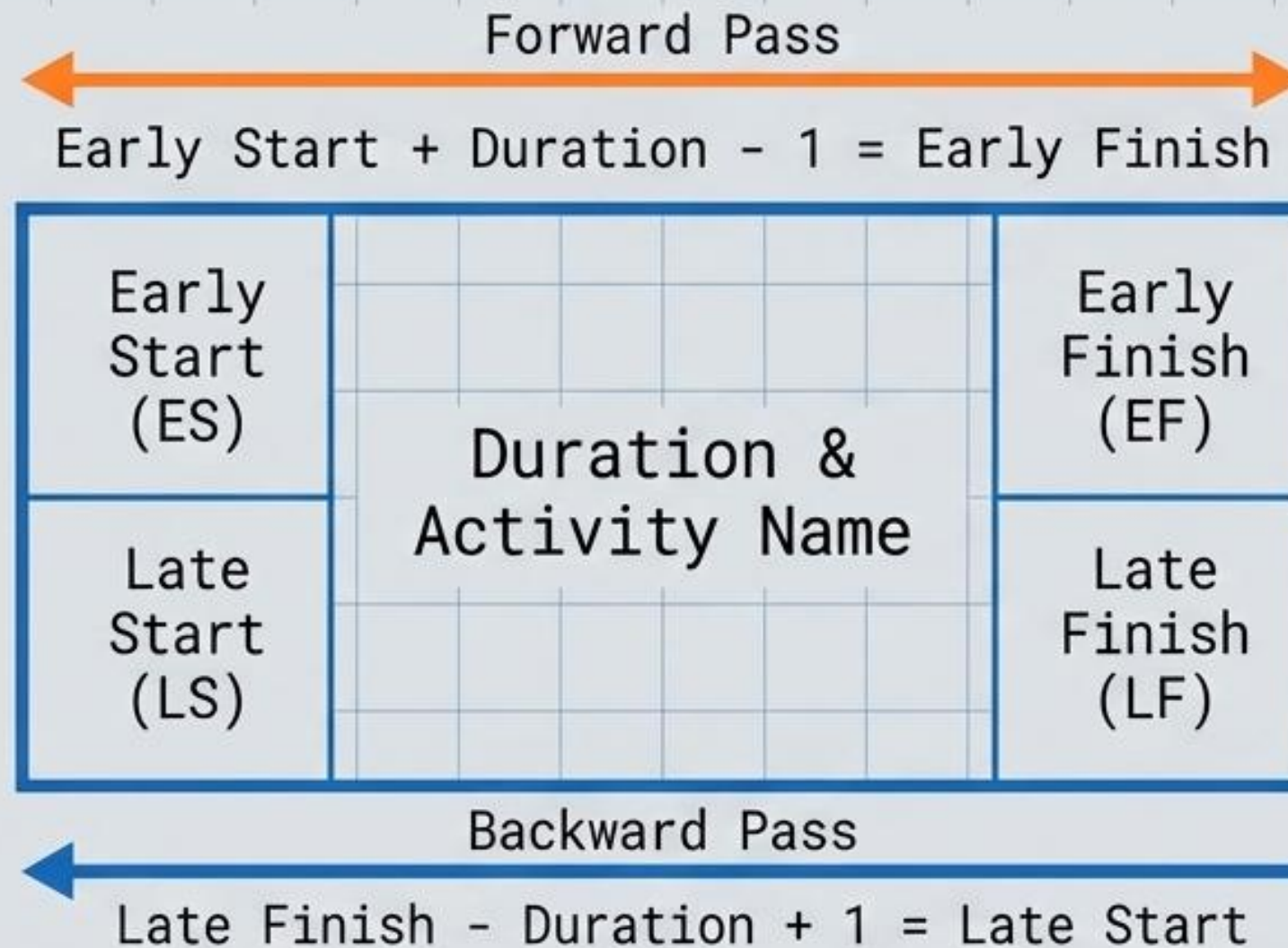
THE LIMITATION

THE THEORETICAL SCHEDULE

Calculates are performed without taking resource limitations into consideration. Resource constraints are applied later.

Note: This analysis uses schedule models alongside other tools like CPM, Critical Chain, and Resource Leveling.

The Anatomy of a Critical Path Node

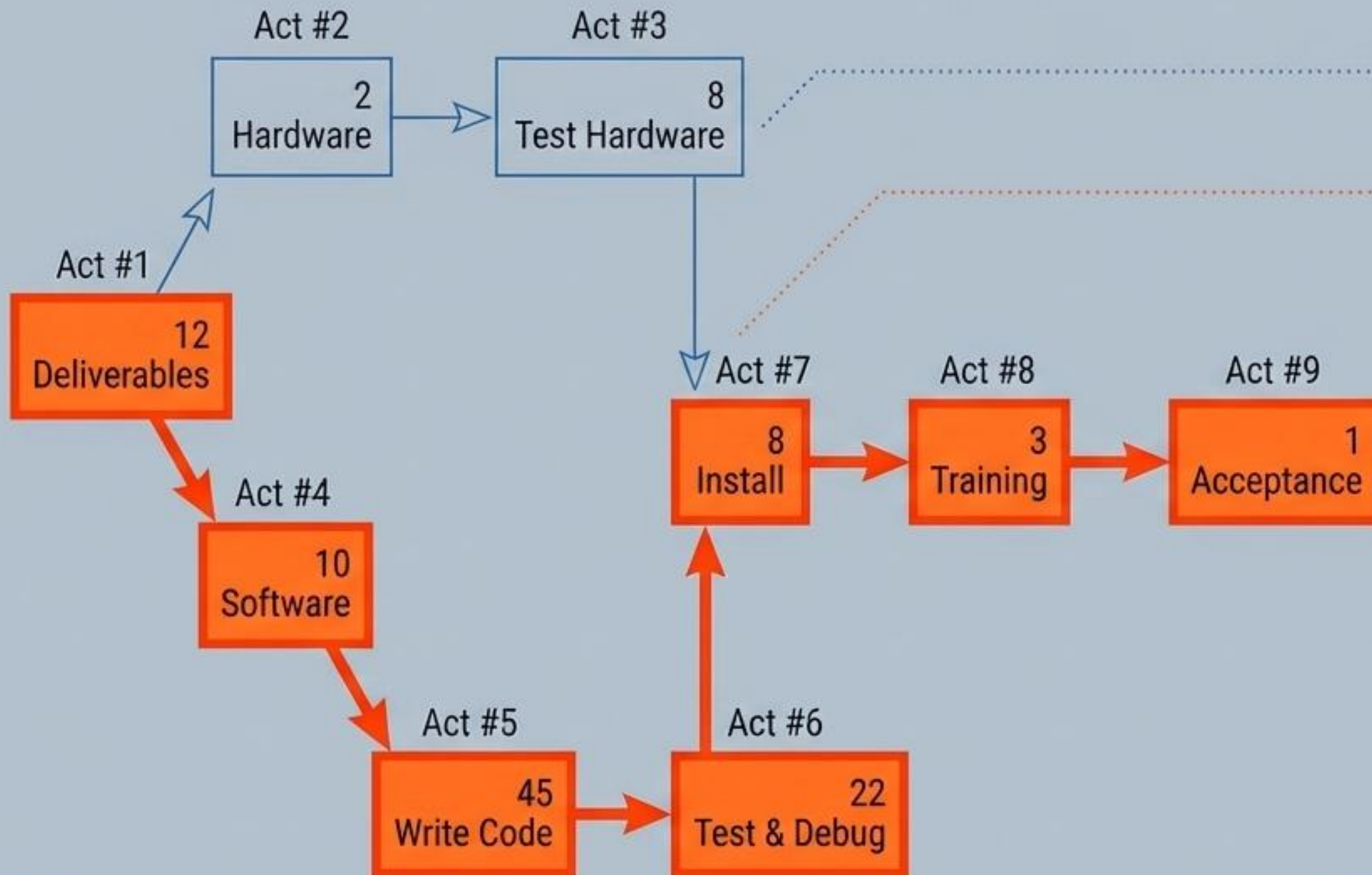


Float = Latest Start - Earliest Start

Total Float (TF): Time delayed without delaying project end.

Free Float (FF): Time delayed without delaying earliest start of successor task.

Tracing the Critical Path (CPM)



Path 1-2-3-7-8-9
= 12+2+8+8+3+1
= 34 days

Path 1-4-5-6-7-8-9
= 12+10+45+22+8+3+1
= 101 days

Core Rule

The Critical Path (CP) is the longest full path on the project, resulting in the minimum project duration.

Float Condition

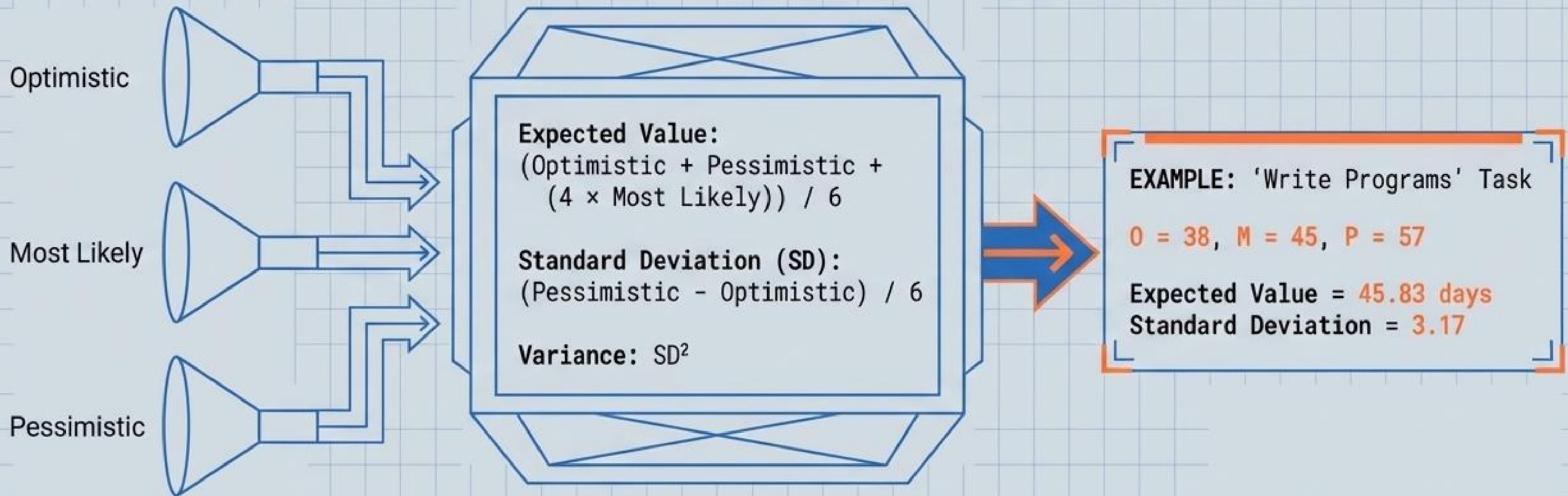
Any task with a float time of 0 (or negative float) is on the critical path. If a task uses up all its float, it becomes a critical path task.

X:000.0
Y:000.0

X:680.0
Y:600.0

PERT: The Mathematical Engine of Uncertainty

CPM uses a single deterministic duration. PERT (Program Evaluation and Review Technique) uses probability and three-point estimates for high-risk, uncertain activities.

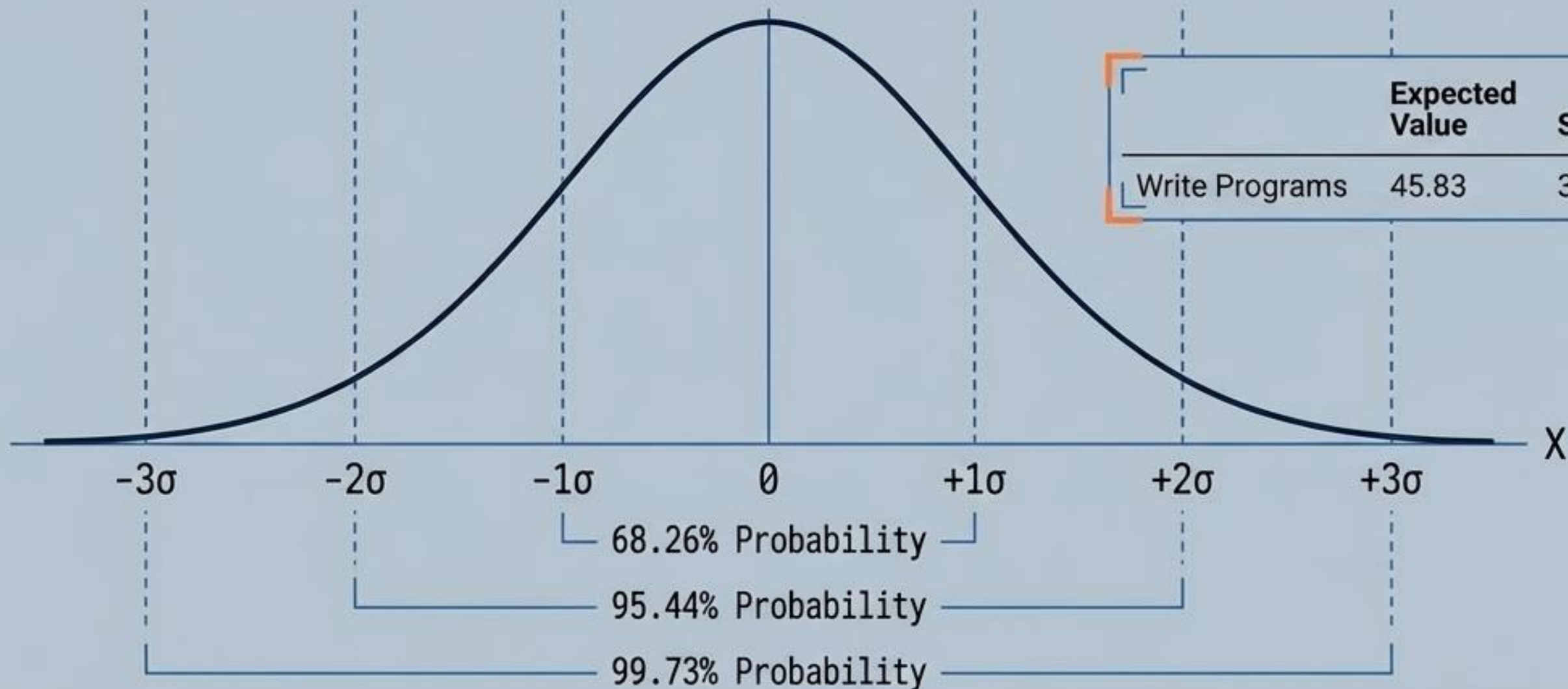


The PERT Confidence Curve



PMP Exam Spotlight

Know the 68% (1 SD) and 95% (2 SD) rules. Higher Standard Deviation = Higher Risk (greater spread between pessimistic and optimistic). You cannot simply add standard deviations across tasks; you must add the variances (SD^2), then find the square root.



	Expected Value	SD	SD squared
Write Programs	45.83	3.17	10.05

X:800.0
Y:000.0

X:680.0
Y:600.0



X:090.0
Y:000.0

X:680.0
Y:600.0



Modifying for Reality: The Critical Chain Method

Built on CPM, but uses nonconservative duration estimates and accounts for limited resource availability. Schedules high-risk tasks early.

Rigid CPM Path

Task A → Task B → Task C → Task D → Task E → Task F → Task G → Task H → Task I → Task J...

Critical Chain Path

Task A

Task B



Task C

Task C

Task E

Task D

Task E

Task F



Feeding Buffers: Added to noncritical paths that feed into the critical chain.

Project Buffers: Added at the very end the critical chain to protect the final deadline.

Feeding Buffers: Added to noncritical paths that feed into the critical chain.

Project Buffers: Added at the very end of the critical chain to protect the final deadline.



PMP Exam Spotlight

CPM manages total float. Critical Chain manages buffer activity durations to protect the schedule from slipping. Tasks are scheduled at their latest possible start/finish dates.

X:680.0
Y:600.0

X:090.0
Y:000.0

X:680.0
Y:600.0

Resource Optimization Diagnostic Matrix

Resource Leveling	Resource Smoothing	Reverse Resource Allocation
Trigger: Resources are overallocated or limited.	Trigger: Under a strict date constraint but need to manage workload.	Trigger: A highly specialized, single resource is needed at a specific exact point.
Action: Adjusts start/finish dates to balance workload. Allocates resources to critical path tasks first.	Action: Modifies activities within their existing float times (e.g., split tasks, reallocate underutilized staff).	Action: Schedules the project in reverse order from the end date backwards.
Impact: Changes the Critical Path and often extends the project end date.	Impact: Does NOT change the Critical Path or project end date.	Impact:

X:000.0 Y:000.0

X:680.0 Y:600.0

Scenario Planning & Timeline Adjustments

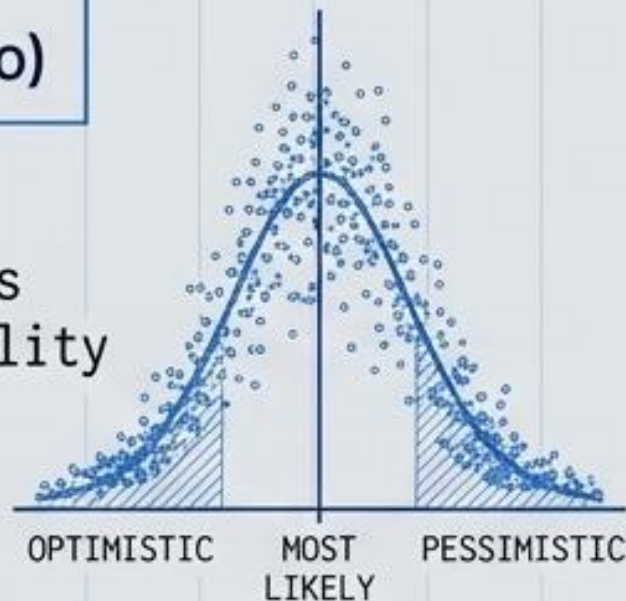
Modeling The Unknown

What-If Scenario Analysis

Testing adverse conditions ("What if a deliverable is delayed?"). Feeds risk response and contingency planning.

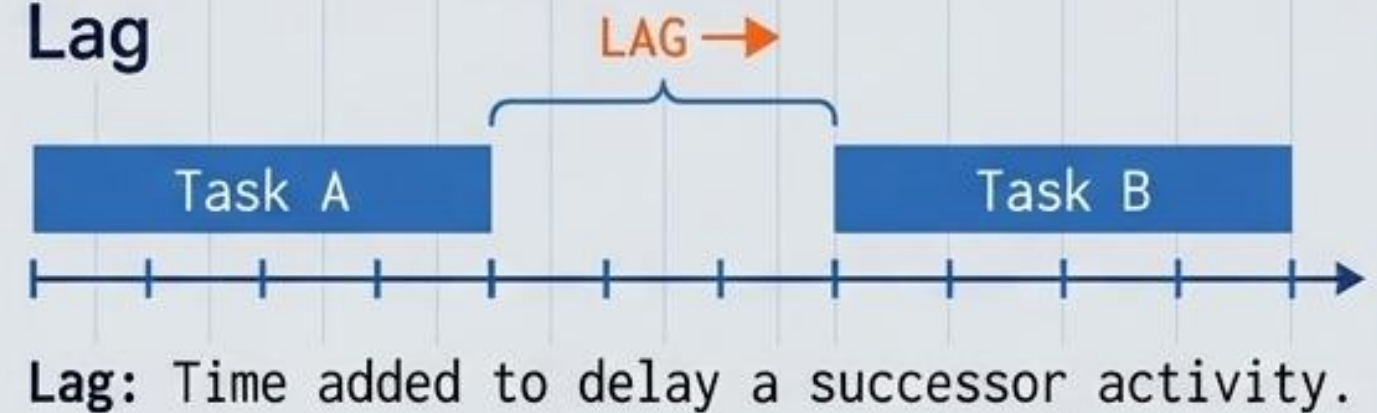
Simulation (Monte Carlo)

Runs probable duration ranges thousands of times to calculate the probability of all possible project completion dates.

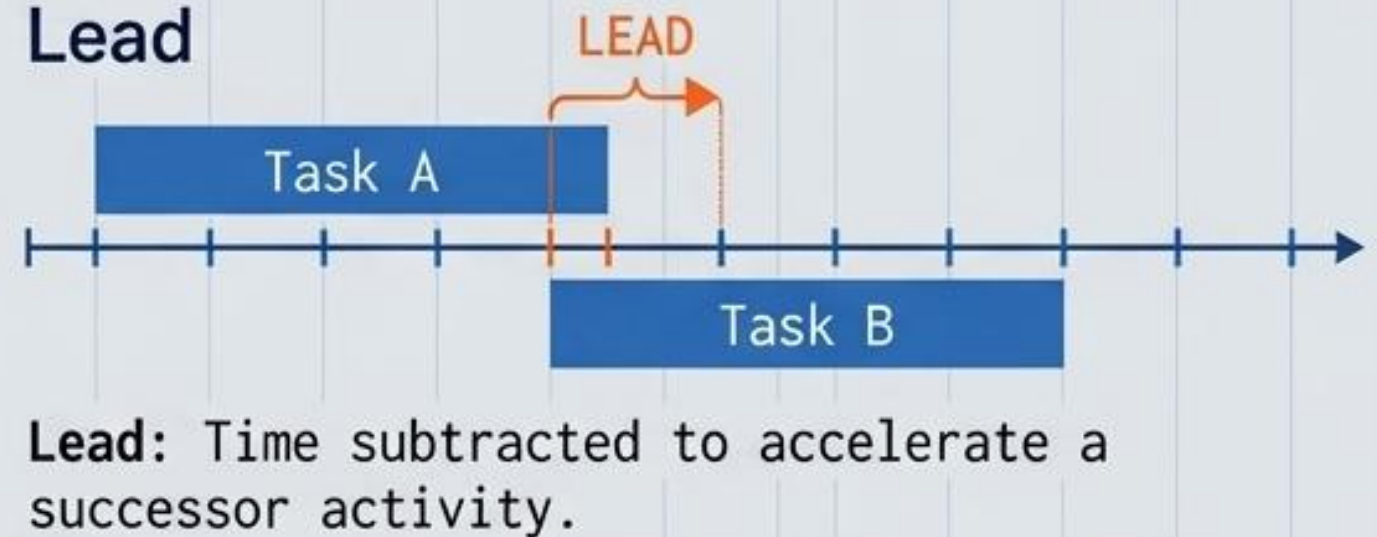


Adjusting Dependencies

Lag



Lead



X:000.0
Y:000.0

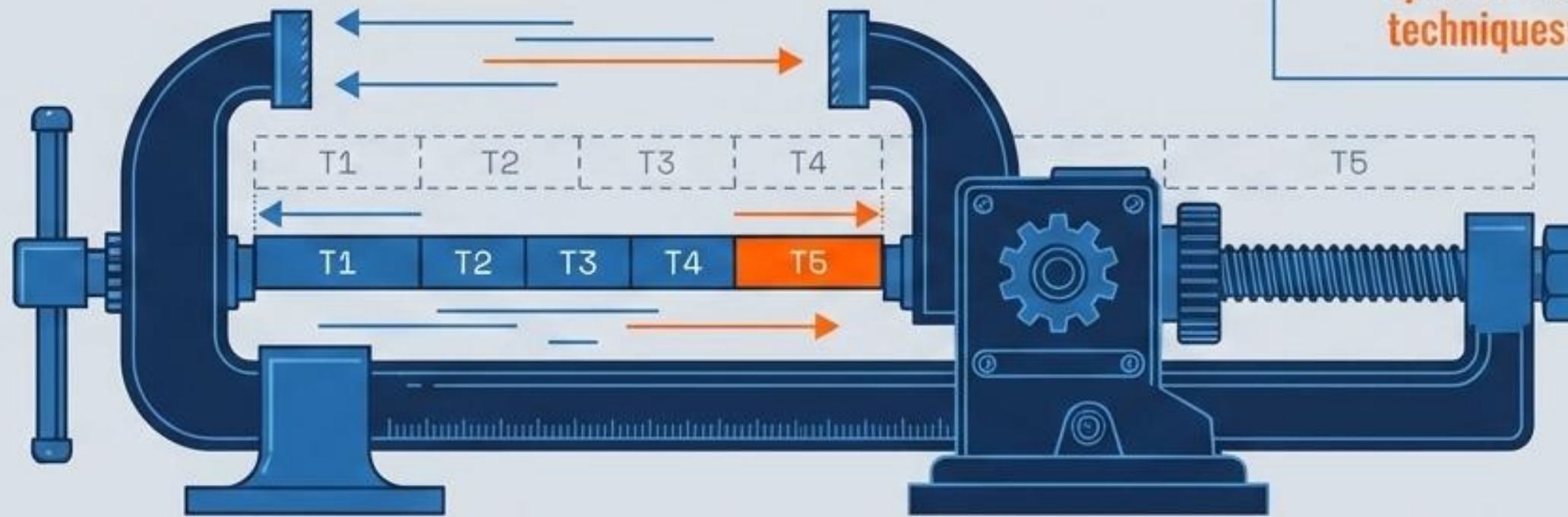
X:680.0
Y:600.0

Schedule Compression: The Emergency Levers



PMP Exam Spotlight

Compression is strictly limited to two specific mathematical analysis techniques: **Crashing** and **Fast Tracking**.



Core Definition

Shortening the project schedule duration to accomplish activities sooner without changing the project scope.

Trigger Condition

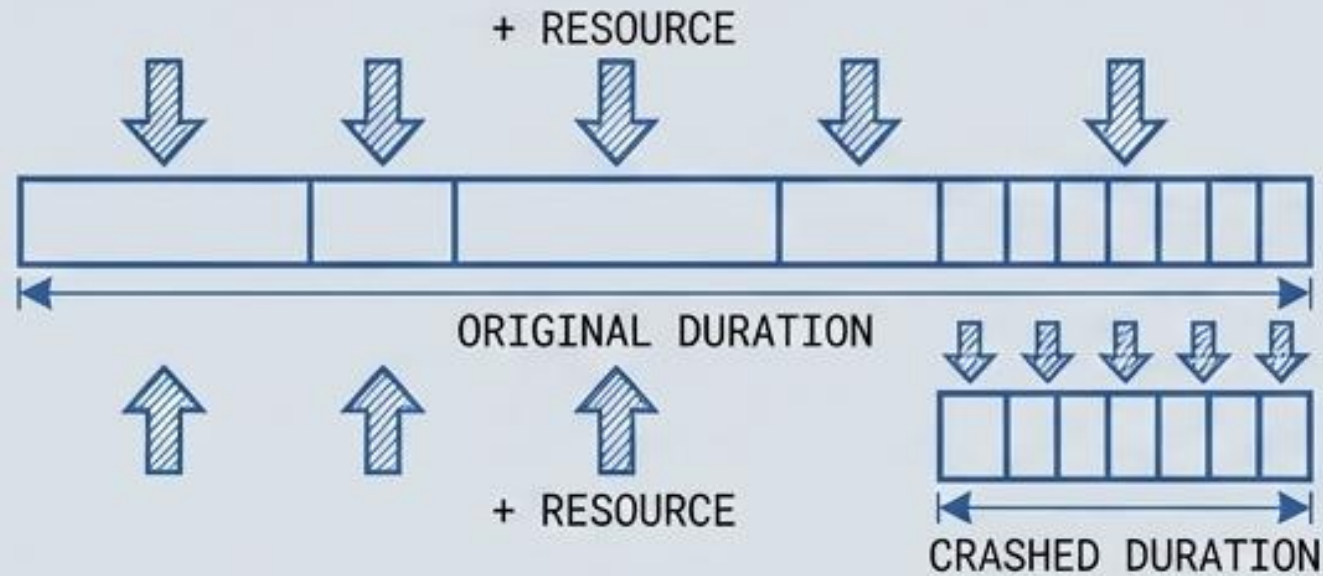
Used when the calculated CPM/PERT end date overshoots a predetermined promised deadline.

X:000.0
Y:000.0

X:680.0
Y:600.0

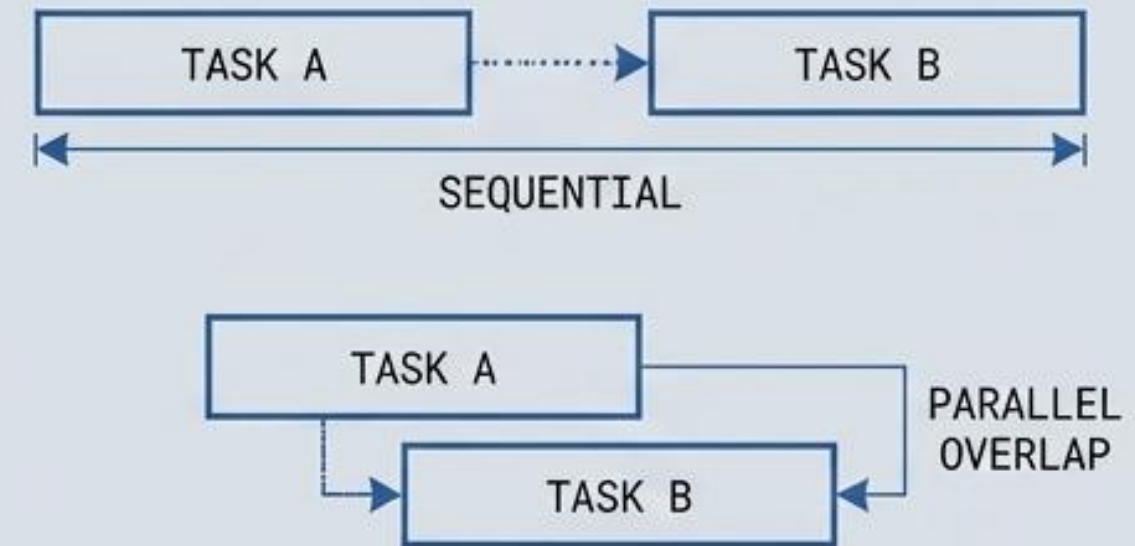
Compression Matrix: Crashing vs. Fast Tracking

Crashing (The Cost Trade-off)



- **Mechanism:** Adding resources (mandatory overtime, expedited shipping) strictly to critical path tasks.
- **Risk:** Increases **Cost** and potentially risk.
- **Rule:** Always check if the critical path has changed after crashing. Seek greatest compression for least cost.

Fast Tracking (The Risk Trade-off)



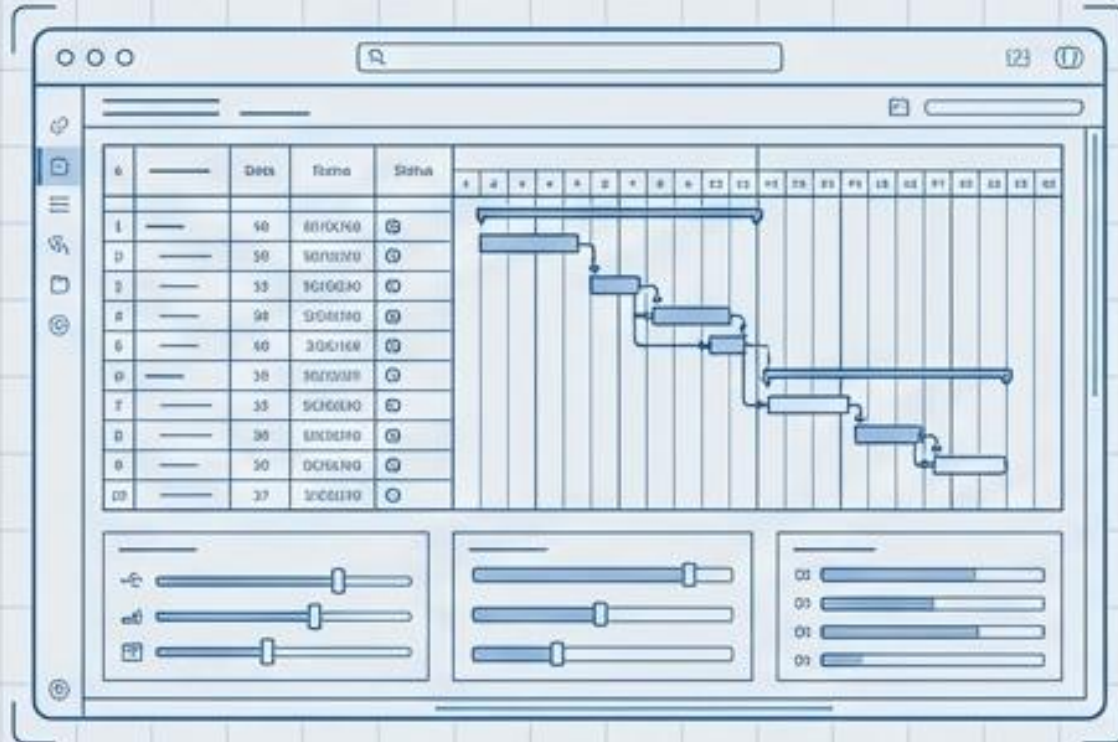
- **Mechanism:** Performing previously sequential tasks/phases in parallel (e.g., coding before design is fully finished).
- **Risk:** Increases **Risk** and potential for rework.
- **Rule:** Only works for activities that can physically overlap (cannot paint before priming).

X:090.0
Y:000.0

X:090.0
Y:000.0

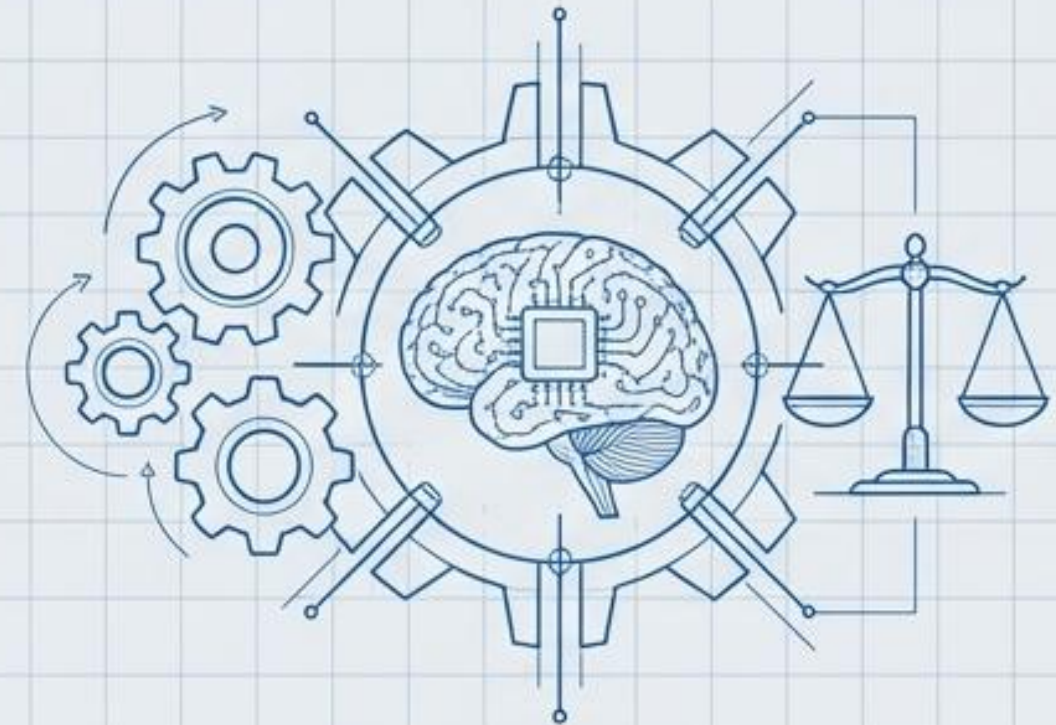
The Algorithm vs. The Project Manager

Software Capabilities



Automates mathematical calculations (forward/backward passes) and handles complex resource-leveling algorithms. Excellent for reporting and stakeholder updates.

The Human Override



Software cannot factor in team relationship dynamics, hidden constraints, or stakeholder expectations.

The Golden Rule

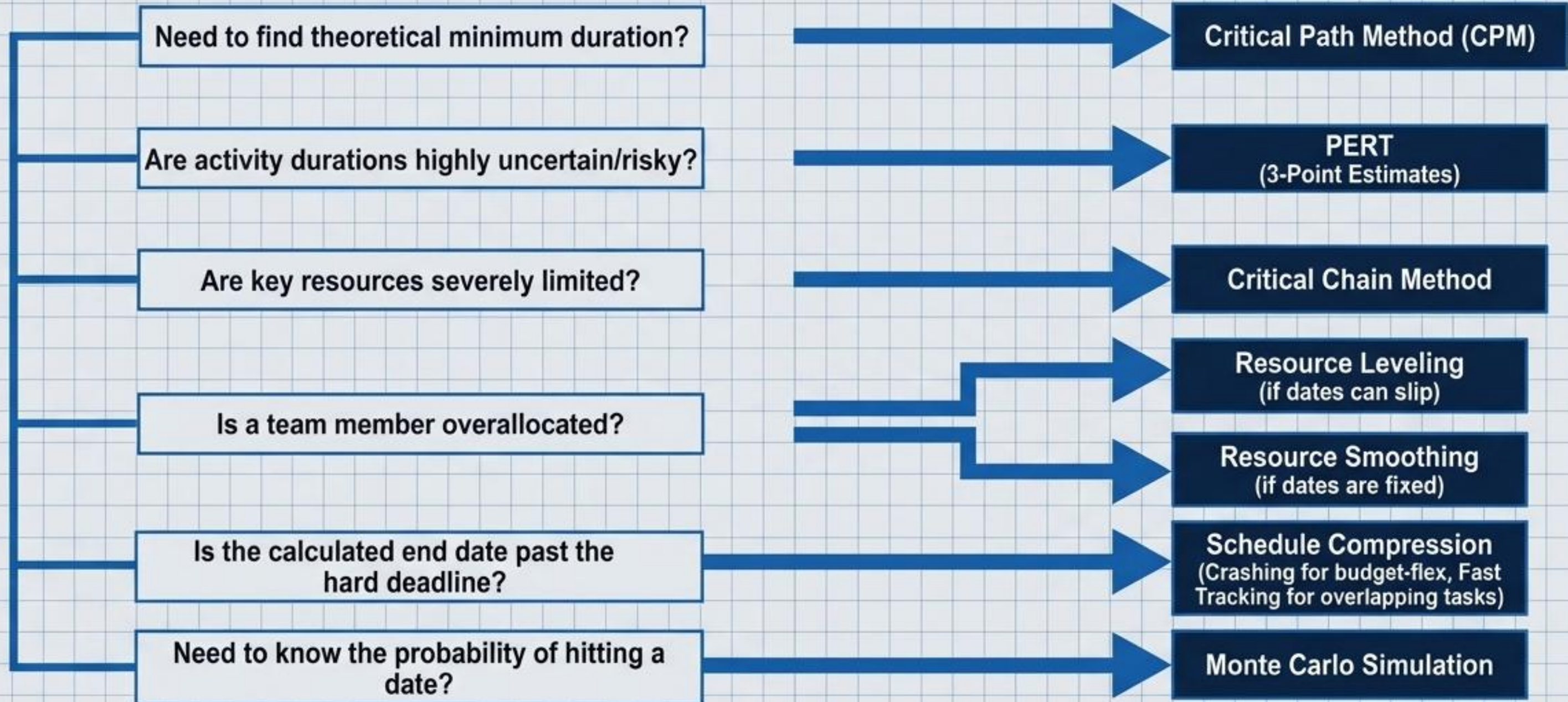
Project management software is a tool, not a substitute for sound PM practices. Never let the software usurp your experience. Don't manage the software instead of the project.

X:090.0
Y:000.0

X:090.0
Y:000.0

NotebookLM

The Master Decision Matrix



A perfectly calibrated schedule balances mathematical theory with real-world constraints.